

The AINUMPSA Resonance Field: A Mathematical Framework for Anomaly Detection in High-Energy Physics and Blockchain-Integrated Systems

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Abstract

This paper introduces the operational mathematical framework of the AINUMPSA system, a novel approach that integrates blockchain tokenomics with high-energy physics data analysis. We define a state space based on token counts, derive a resonance tensor, and present experimental predictions for the Large Hadron Collider (LHC) at 120 GeV. The framework predicts specific anomaly patterns, including a resonance frequency and angular distributions, which can be tested against Standard Model predictions.

1 Part I: The State Space and Resonance Tensor

1.1 Postulate 1: The $1 > 0$ / $0 < 1$ Token Space

We introduce a real (non-complex) state space, as measurements deal with counts, not probability amplitudes.

Let B_+ be the number of tokens on the " $1 > 0$ " blockchain, and B_- be the number of tokens on the " $0 < 1$ " blockchain. The state of the system is defined as a vector:

$$S = \begin{bmatrix} B_+ \\ B_- \end{bmatrix}, \quad B_+, B_- \in \mathbb{R}^+ \quad (1)$$

The normalization condition, representing a conserved total information supply, is:

$$B_+ + B_- = N_{\text{total}} = \text{constant} \quad (2)$$

Here, B_+ is identified with positive energy (matter, future time) and B_- with negative energy (antimatter, past time). Their constant sum is a fundamental information conservation law.

1.2 The Resonance Tensor

We define the AINUMPSA resonance tensor $\mathbf{T}$ not as a metric, but as an operational vector field. For a given spacetime point (x, t) , we define a scalar ' $S(x, t)$ ', and the tensor components are given by:

$$T(x, t) = \alpha \cdot S(x, t) + \beta \cdot \nabla^2 S(x, t) + \gamma \cdot \frac{\partial S}{\partial t}(x, t) \quad (3)$$

Here, α, β, γ are system constants to be calibrated with LHC data. ∇^2 is the spatial Laplacian, and $\frac{\partial}{\partial t}$ is the time derivative.

The equilibrium condition, representing perfect resonance, is:

$$T_+ + T_- = 0 \quad (4)$$

2 Part II: Experimental Predictions for the LHC

2.1 Prediction 1: Resonance Frequency

For collision energies of $E = 120 \text{ GeV}$, we predict the emergence of "nests" (anomalies) when the following condition is met:

$$E \cdot \langle T \rangle = \hbar \omega_{\text{res}} \quad (5)$$

From the AINUMPSA anomaly map, the average tensor value is $\langle T \rangle \approx 0.70$. Thus, the predicted resonance frequency is:

$$\omega_{\text{res}} = \frac{120 \text{ GeV}}{\hbar} \cdot 0.70 \approx 1.28 \times 10^{26} \text{ Hz} \quad (6)$$

This corresponds to a wavelength of approximately $2.34 \times 10^{-18} \text{ m}$, placing it in the hard gamma-ray spectrum. The Standard Model predicts no spectral lines in this region, making this a key falsifiable prediction.

2.2 Prediction 2: Angular Locations of Anomalies

From the anomaly map, we identify specific angular directions where the tensor T deviates from the average:

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ \quad (7)$$

For each of these angles, we predict a local increase in the scalar field $\Phi(x, t)$ of:

$$\Delta\Phi \approx \pm 0.05 \cdot T_{\text{avg}} \quad (8)$$

This prediction can be tested by comparing LHC data for collisions under these angles against data from other directions. The difference is expected to be statistically significant (above 3σ).

2.3 Prediction 3: The "Nests" Equation

Nests are defined as points (x_0, t_0) where the energy difference between reality and its mirror reflection vanishes. We define a scalar field:

$$\Phi(x, t) = T_+(x, t) - T_-(x, t) \quad (9)$$

The nests satisfy the following equation with a specific boundary condition:

$$\frac{\partial^2 \Phi}{\partial t^2} - c^2 \nabla^2 \Phi = -\kappa \cdot \epsilon_{\mu\nu\rho\sigma} \frac{\partial J^\rho}{\partial x_\sigma} \quad (10)$$

The boundary condition for standing waves is:

$$\Phi(x = 0, t) = 0 \quad \text{and} \quad \Phi(x = L, t) = 0 \quad (11)$$

where L is the size of the system (e.g., the LHC detector diameter). The solution yields standing waves:

$$\Phi(x, t) = A \sin(kx) \cos(\omega t) \quad (12)$$

With $k = \frac{n\pi}{L}$ and $\omega = c \cdot k$. The nests are located at the nodes of these standing waves:

$$x = \frac{m\pi}{k} = \frac{mL}{n}, \quad \text{where } m = 0, 1, 2, \dots \quad (13)$$

Acknowledgments

This work was developed as part of the AINUMPSA project, an autonomous information system integrating blockchain, physics simulation, and real-time analysis.

Data and Code Availability

All code and data used in the AINUMPSA project are available in the public GitHub repository and are archived on Zenodo. The workflow is designed to be fully reproducible.

A Appendix: Data Collection Algorithm for LHC

1. For each collision event, record: E (energy), t (time), θ and ϕ (angles in lab frame).
2. Record B_+ and B_- (the corresponding token counts from the blockchain at that moment).

3. Calculate the local resonance tensor:

$$T_+ = \alpha B_+ + \beta \nabla^2 B_+ + \gamma \frac{dB_+}{dt} \quad (14)$$

$$T_- = \alpha B_- + \beta \nabla^2 B_- + \gamma \frac{dB_-}{dt} \quad (15)$$

4. Calculate the difference:

$$\Phi = T_+ - T_- \quad (16)$$

5. Check the nest condition:

$$\text{if } |\Phi| < \epsilon \text{ and } |\nabla \Phi| < \epsilon, \text{ save coordinates to gniazda.json} \quad (17)$$

6. Update the Attractor A :

$$A_{n+1} = A_n + \gamma \cdot (1 - A_n) \cdot N_{\text{gniazda}} \quad (18)$$